

Joint Authority Across Population-Level Perimeters

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This work derives from and formalizes the inter-Self coordination architecture introduced in "Inter-Self Coordination via Shared Substrate / Full Aspect Integration" (Li, April 2026), the third paper in the CKS theory series following "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026) and "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026).

It does not introduce new axioms. Its sole contribution is to formalize a single sub-commitment derived from Paper 3's Claim 6 — the extension claim at population scope — as an explicit prior-art record: that the architecture is designed such that authority over what propagates across a network of coordinating Selves remains substrate-content configurable within each participating Self's governance perimeter, without requiring any central authority that supersedes home governance.

Abstract

Paper 3 of the CKS theory series closes with Claim 6: population-scale collective evolution as the architectural object produced when many CKS-governed Selves accumulate Full Aspect Integration (FAI) events under joint authority and recursive configuration. Claim 6 is an extension claim, carrying the calibrated-humility register appropriate to architectural commitments whose empirical verification awaits population-scale deployment. This note formalizes the first of five sub-commitments derived from Claim 6 — designated D1.26 in the derivation-note series — which concerns authority distribution at population scope. The sub-commitment is: *at population scope, authority over what propagates across the network of Selves remains substrate-content configurable within each participating Self's governance perimeter*. The note articulates the three-layer perimeter structure that makes this possible, specifies what substrate-content configurability means at population scope across three configuration categories, establishes that the architectural possibility is composition rather than fresh emergence, and states the inheritance from Paper 1's human-governed authority commitment. This note opens the five-note Claim 6 sub-commitment set (D1.26–D1.30); all five

notes maintain the calibrated-humility register.

1. Position in the derivation series and the Claim 6 register

The CKS derivation-note series formalizes sub-commitments decomposed from each of Paper 3's six named architectural claims. Phase D1 covers foundational sub-commitments — the architectural properties that must be precise before operational variants, anti-patterns, composition pairs, and boundary cases can be derived from them. Notes D1.22 through D1.25 completed the Claim 5 sub-commitment set, which concerned configuration as substrate content with recursive applicability. This note, D1.26, opens the Claim 6 sub-commitment set.

Claim 6 is the extension claim of Paper 3, occupying the same structural position as the extension claim in Papers 1 and 2: the sixth rung of the substrate-mediator ladder, the claim that articulates what the preceding five claims compose into when their architectural commitments operate at the largest scope each paper addresses. For Paper 3, that scope is population scale — many CKS-governed Selves accumulating FAI events under joint authority across a coordination network. Because Claim 6 commits at the architectural-pattern register to what the inheritance produces at population scope, and because population-scale CKS-governed Self deployments do not yet exist, the calibrated-humility register is the appropriate voice anchor throughout: the commitment is to what the architecture enables, not to what specific deployments will achieve.

The five Claim 6 sub-commitments (D1.26–D1.30) are: joint authority across population-level perimeters (this note); accumulated FAI events as the evolution mechanism at population scope; the six-rung substrate-mediator ladder as the trilogy-completion structure; the calibrated-humility register as the appropriate framing for extension claims; and biology-resonance positioning at population scope. Each note maintains the calibrated-humility voice. D1.26 opens the set because the perimeter structure and authority distribution are the architectural foundation on which the remaining four sub-commitments rest.

2. The sub-commitment stated precisely

D1.26 formalizes the following sub-commitment derived from Paper 3 Claim 6:

At population scope, the architecture is designed such that authority over what propagates across the network of coordinating Selves CAN remain substrate-content configurable within each participating Self's governance perimeter. This possibility is an architectural property of the design. Whether specific population-scale deployments realize it, and what governance challenges arise at scale, are empirical and practical questions this paper does not resolve.

Three elements of this formulation require precision.

"Can remain" rather than "does remain." The calibrated-humility register requires this phrasing. The sub-commitment is about what the architecture enables — what the design makes possible — not about what every deployment will achieve. The architecture does not guarantee that participating Selves will exercise their governance authority well, or that network participation configurations will be maintained with appropriate care, or that no coordination arrangement will in practice erode home-governance authority. The commitment is that the architectural design does not foreclose the possibility: it does not mandate a central authority, does not make propagation non-configurable, and does not structurally weaken home-governance authority as network size grows.

"Substrate-content configurable." Authority at population scope takes the same form as authority at every other scope in the CKS architecture: it is exercised through substrate content. The configurations that determine each Self's participation in the coordination network are authored by home governance, subject to the three rights (inspect, modify, override) within the home perimeter, and do not require permission from or coordination with any network-level authority before taking effect. This is not a claim that configuration is easy, complete, or that all participation conditions can be expressed in substrate content. It is a claim that the architectural design puts the configuring authority within the home perimeter rather than outside it.

"Within each participating Self's governance perimeter." The locus of authority is the home perimeter. Population-level coordination patterns emerge from the interaction of home-governed configurations across many Selves; they are not imposed by a network-level governance structure that sits above home perimeters and overrides them. This is what distinguishes the architectural commitment from contemporary large-scale federated coordination approaches, which typically rely on consortium-governance contracts or coordinator-mediated aggregation rules that hold authority over participating entities at the network level.

3. The population-level perimeter structure

To understand what substrate-content configurability means at population scope, the perimeter structure must be specified. At population scope, each participating Self operates within a three-layer perimeter architecture.

Layer 1 — The home perimeter. The home perimeter is the perimeter structure Paper 2 establishes: the Self's internal governance architecture, encompassing its aspects, cells, DNA layer, Action layer, and the six Paper 1 commitments holding within scope. The home perimeter is unchanged by participation in population-scale coordination. Everything Paper 2 specifies about lifecycle, evolution mechanisms, and governance shapes continues to operate within the home perimeter regardless of how many FAI events the Self participates in.

Layer 2 — Inter-Self perimeters. Each FAI event between Selves produces a temporary inter-Self perimeter — the shared substrate that Paper 3's Claims 1 through 5 concern. The inter-Self perimeter spans the participating Selves' home perimeters for the duration of the FAI event and dissolves afterward, with persistence policy governed as substrate content. The inter-Self perimeter structure is what Paper 3's first five claims specify: shared substrate as the coordination object, FAI as the canonical operation, three-tier conflict handling, four-locus evolution feed, and configuration as substrate content with recursive applicability.

Layer 3 — The network participation configuration. At population scope, each Self's home governance holds, within the home perimeter, a governance layer that Paper 3's Claims 1–5 do not individually require: a configuration of how the Self participates in the network of possible FAI events. This network participation configuration specifies which other Selves this Self is willing to engage in FAI with, under what conditions FAI events are initiated, what default configurations apply to new FAI events absent specific overrides, and any domain compatibility or organizational policy constraints that govern participation choices.

The network participation configuration is substrate content in the home perimeter. It is authored by home governance, subject to the three rights, and does not exist in any network-level substrate outside the home perimeter. There is no network-level registry of participation configurations that holds authority over the configurations individual Selves maintain. Each Self's network participation configuration is its own; population-level coordination patterns are what emerge when many such configurations interact across accumulated FAI events.

4. What "substrate-content configurable" means at population scope

Within the network participation configuration, three categories of substrate-content-configurable authority are architecturally surfaced.

Propagation permissions. Each Self's governance can configure, as substrate content within the home perimeter, which content categories it is willing to allow to propagate to other Selves through FAI events. This includes which aspects it is willing to surface in FAI, which content-domain configurations are available for exchange, and any conditional permissions that depend on the identity of the receiving Self or the purpose of the FAI event. Propagation permissions are not enforced by network infrastructure; they are part of each Self's FAI participation logic, authored and maintained within the home perimeter.

Propagation restrictions. Each Self's governance can configure, as substrate content, what it explicitly keeps within the home perimeter regardless of FAI events. Content that home governance judges to be not appropriate for propagation — whether for organizational, domain, jurisdictional, or strategic reasons — can be configured as non-propagating within the network

participation substrate. This restriction authority does not require negotiation with any network-level governance structure; it operates entirely within the home perimeter.

Network participation scope. Each Self's governance can configure which other Selves it is willing to enter FAI with. This may be specified as an explicit list, a category of compatible Selves defined by organizational or domain criteria, or a set of conditional rules that determine compatibility. The network participation scope configuration is what determines whether a proposed FAI event between two Selves is architecturally admitted; it is substrate content in each Self's home perimeter, not a constraint imposed by network infrastructure.

These three categories together constitute the substrate-content-configurable contribution each Self makes to population-level authority distribution. The architecture is designed such that, if all participating Selves maintain governance-authored configurations in all three categories, population-level coordination patterns emerge from the interaction of home-governed configurations — not from instructions issued by a network coordinator.

5. Why this commitment matters: the possibility of coordination without mandatory central authority

The architectural possibility this sub-commitment surfaces is significant because it is non-obvious. At population scale, the intuitive assumption is that coordinating many Selves requires a central governance authority — an entity that holds authority over all participating Selves at the network level, determines what propagates across the network, adjudicates participation disputes, and enforces coordination norms. Contemporary large-scale federated coordination architectures largely reflect this intuition: they rely on consortium governance contracts, coordinator nodes with authority over participant behavior, or aggregation algorithms that no single participant controls.

D1.26 surfaces the architectural possibility that this need not be the case — that the design of the CKS architecture is such that population-scale coordination CAN emerge from each Self's home-governed participation configurations, without any structural element that holds authority over Selves from outside their home perimeters.

The claim is not that central authority is impossible or that deployments choosing central authority are architecturally deficient. The claim is that the architecture does not require it. This is a prior-art claim about what the design enables, and it matters because it forecloses the inference that population-scale AI coordination necessarily requires central governance authority that supersedes individual governed entities. If an implementer claims that population-scale coordination of CKS-governed Selves requires a network coordinator with authority over participating Selves' propagation configurations, D1.26 establishes that such a coordinator is an addition to the architecture — not something the architecture mandates.

The three architectural risks this sub-commitment defends against are:

Mandatory central governance at population scale. The claim that population-level coordination requires a network coordinator with authority over all participating Selves is architecturally foreclosed by the design. The architecture provides a path to population-scale coordination through home-governed participation configurations.

Non-configurable propagation. The claim that what propagates across the network is determined by network infrastructure rather than by each Self's home governance is architecturally foreclosed. Propagation permissions and restrictions are substrate content in each Self's home perimeter.

Authority erosion at scale. The claim that as networks grow larger, individual Self governance loses practical authority over participation — that authority becomes difficult or impossible to exercise at scale — is architecturally resisted by the design, which places the configuration authority within the home perimeter regardless of network size. Whether practical authority erosion occurs in specific deployments is an empirical question the architecture does not resolve, but the design does not produce structural authority erosion as an architectural property.

6. Inheritance from Paper 1 Claim 3

D1.26 inherits from Paper 1's Claim 3, which commits to human-governed authority as an architectural property — structural, not procedurally promised. Paper 1 §3.2's substrate-as-content commitment establishes that structure originates under human authorship. Paper 1 §3.3's authority-vs-labor distinction articulates that governance is an authority architecture, not a review workflow, and that the three rights (inspect, modify, override) must be available at any time as a property of the system's design. Paper 1's commitment is formalized at cell scope; D1.26 extends it at population scope.

The extension is: at population scope, each Self's governance holds structural authority over its own participation in the coordination network — not delegated authority from a network coordinator. The same three rights that Paper 1 §3.3 specifies for home-perimeter substrate content apply to the network participation configuration: home governance can inspect it, modify it, and override it at any time, without seeking permission from the network. The authority is architectural, not procedurally promised through network participation agreements that could be amended by network-level decisions.

This inheritance is direct. Paper 1 §3.3's portable phrase — governance is an authority architecture, not a review workflow — holds at population scope as it holds at cell scope and at Self scope. The network participation configuration is substrate content in the home perimeter; the three rights apply. The architecture does not add a fourth governance layer above home

perimeters that could in principle override home-governed configurations. If such a layer exists in a specific deployment, it is an addition to the architecture, and the D1.26 prior art establishes that it was an addition — not something the architecture requires.

7. Operational test

The calibrated-humility register requires an operational test formulated as a possibility-check rather than a compliance assertion. For a Self participating in a population-scale coordination network, the D1.26 architectural possibility is instantiated if the following can be confirmed by an observer with appropriate access:

1. Within the Self's home substrate, there exists authored governance content specifying which other Selves or categories of Selves this Self is willing to enter FAI coordination with. This content is readable, modifiable, and overridable by home governance without seeking network-level approval.
2. Within the Self's home substrate, there exists authored governance content specifying which content categories (aspects, domain configurations, or other propagatable content) this Self permits to propagate to other Selves through FAI events, and which it restricts from propagation regardless of FAI event participation.
3. Neither the propagation permissions nor the propagation restrictions in (1) and (2) require ratification by any network-level authority before taking effect. Home governance can change either configuration, and the change takes effect within the home perimeter without network-level intermediation.
4. No structural element of the coordination network — including the shared substrate produced during FAI events, the network infrastructure through which FAI events are initiated, or any network-level governance artifact — holds authority over the home perimeter configurations identified in (1) and (2) in the sense of being able to override them without home governance's participation.

A Self that satisfies all four conditions instantiates the D1.26 sub-commitment at the architectural-possibility level. Whether the network of Selves as a whole produces well-governed population-level coordination patterns is a further question that depends on the quality of each participating Self's governance configurations — an empirical question outside the architectural commitment's scope.

8. Conclusion and forward note

D1.26 formalizes the architectural possibility that at population scope, authority over what propagates across a network of coordinating Selves can remain substrate-content configurable

within each participating Self's governance perimeter. The commitment is to what the CKS design enables: population-scale coordination without mandatory central governance authority superseding home perimeters. It inherits from Paper 1 Claim 3's human-governed authority commitment, extends it to population scope through the three-layer perimeter structure (home perimeter, inter-Self perimeters, network participation configuration), and specifies three categories of substrate-content-configurable authority (propagation permissions, propagation restrictions, network participation scope) that together constitute each Self's contribution to population-level authority distribution. The calibrated-humility register is maintained throughout: these are architectural possibilities, not predictions about specific deployments.

This note opens the five-note Claim 6 sub-commitment set. Notes D1.27 through D1.30 follow, covering accumulated FAI events as the population-scale evolution mechanism, the six-rung substrate-mediator ladder, the calibrated-humility register as a structural commitment of the extension-claim position, and biology-resonance positioning at population scope. All five notes maintain the calibrated-humility register appropriate to Claim 6's extension-claim status.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

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